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Kinematics 6 HW

$$1a) r = \begin{pmatrix} t^2 \\ 5t - 2t^2 \end{pmatrix}$$

$$a = \begin{pmatrix} 2 \\ -4 \end{pmatrix} = 2i - 4j$$

b) If particle at rest, $v = 0$ for some t

$$\begin{pmatrix} 2t \\ 5 - 4t \end{pmatrix} = 0$$

$$\text{when } t = 0, v = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

when $t = \frac{5}{4}$, $v = \begin{pmatrix} 2.5 \\ 0 \end{pmatrix}$ $\therefore$ particle never at rest

$$2) a = \begin{pmatrix} 2t \\ 3 \end{pmatrix}$$

$$v = \begin{pmatrix} t^2 + 1 \\ 3t - 2 \end{pmatrix}$$

$$s = \begin{pmatrix} \frac{1}{3}t^3 + t + 2 \\ \frac{3}{2}t^2 - 2t \end{pmatrix}$$

$$3) s = \begin{pmatrix} 2t + 1 \\ -3t^2 + 1 \end{pmatrix}$$

$$s = \begin{pmatrix} t^2 + t + c_1 \\ -t^3 + t + c_2 \end{pmatrix}$$

$$= \begin{pmatrix} t^2 + t \\ -t^3 + t \end{pmatrix} \quad (\text{at } t=0, s = \begin{pmatrix} 0 \\ 0 \end{pmatrix})$$

$$\therefore c_1 = 0 \text{ or } c_2 = 0$$

When $t = 3$,

$$s = \begin{pmatrix} 12 \\ -24 \end{pmatrix}$$

$$|s| = \sqrt{144 + 576} \\ = 26.8$$

$$4) F = \begin{pmatrix} 4t \\ 6 \end{pmatrix}$$

$$F = ma \quad m = 2$$

$$a = \begin{pmatrix} 2t \\ 3 \end{pmatrix}$$

$$v = \begin{pmatrix} t^2 + c_1 \\ 3t + c_2 \end{pmatrix} = \begin{pmatrix} t^2 \\ 3t + 9 \end{pmatrix} \quad \text{since initial velocity is } 0, \quad \begin{matrix} 4t^2 + c_1 = 0 \therefore c_1 = 0 \\ 12t + c_2 = 9 \therefore c_2 = 9 \end{matrix} \quad (t=0)$$

$$\text{when } t = 3, v = \begin{pmatrix} 9 \\ 14 \end{pmatrix}$$

9) If particle at origin, when $r_x = 0$, $r_y = 0$.

$$\text{When } r_x = 0, t = 1 \text{ or } -1.$$

$$\begin{matrix} \downarrow & \downarrow \\ r_y = -2 & r_y = 4 \end{matrix}$$

So particle is never at the origin.

5 ii) When moving parallel to y-axis,
 $\uparrow$ component = 0.

$$\underline{v} = \begin{pmatrix} -2t \\ 2t-3 \end{pmatrix}$$

$$-2t = 0 \therefore t = 0$$

$$\text{iii) } \underline{s} = \begin{pmatrix} 1-t^2 \\ t^2-3t \end{pmatrix}$$

If vectors $\perp$, $\text{grad. of } \underline{s} \times \text{grad. of } \underline{v} = -1$.

$$\left(\frac{1-t^2}{1-t^2} \times \frac{2t-3}{-2t} \right) = -1$$

$$\frac{2t^3 - 4t^2 + 4t}{2t^3 - 2t} = -1$$

$$4t^3 - 4t^2 + 4t = 0$$

$$t(4t^2 - 4t + 4) = 0$$

$$t \neq 0 \text{ so } 4t^2 - 4t + 4 = 0$$

$4t^2 - 4t + 4$ does not factorize:

$$b^2 - 4ac = 81 - 4 \times 4 \times 7 = 81 - 112 = -31$$

Since $b^2 - 4ac < 0$, expression has no real roots
 $\therefore$ particles are never perpendicular

iv) If ever parallel, $\text{grad. of } \underline{s} = \text{grad. of } \underline{v}$

$$\frac{1-t^2}{1-t^2} = \frac{2t-3}{-2t}$$

$$-2t^3 + 6t^2 = 2t - 3 - 2t^3 + 3t^2$$

$$3t^2 - 2t + 3 = 0$$

$$\text{Here, Discriminant } b^2 - 4ac = 4 - 4 \times 3 \times 3 = -32$$

$\therefore 3t^2 - 2t + 3 = 0$ has no solutions

So $\underline{s}$ and $\underline{v}$ are never parallel

$$\begin{aligned} \text{v) distance to origin } d &= \sqrt{(1-t^2)^2 + (t^2-3t)^2} \\ &= \sqrt{t^4 - 2t^2 + 1 + t^4 - 6t^3 + 9t^2} \\ &= \sqrt{2t^4 - 6t^3 + 7t^2 + 1} \end{aligned}$$

So minimum distance is when $t = 0$.

Exam Q5

$$\text{a) } A: \begin{pmatrix} t^2 \\ 3t-1 \end{pmatrix} \quad v = \begin{pmatrix} 2t \\ 3 \end{pmatrix}$$

$$B: \begin{pmatrix} 1-2t^2 \\ 3t-2t^2 \end{pmatrix} \quad v = \begin{pmatrix} -4t \\ 3-4t \end{pmatrix}$$

$$2t^2 + 9 = 16t^2 + 16t^2 - 24t + 9$$

$$30t^2 - 24t = 0$$

$$t = 0, \text{ and } t = \frac{30}{24} = \frac{5}{4}$$

$$\text{b) } A \text{ to } B = B - A = \begin{pmatrix} 1-3t^2 \\ -2t^2+1 \end{pmatrix}$$

$$\begin{aligned} d^2 &= 9t^4 - 6t^2 + 1 + 4t^4 - 4t^2 + 1 \\ &= 13t^4 - 10t^2 + 2 \end{aligned}$$

c) shortest distance:

$$\sqrt{13t^4 - 10t^2 + 2} = 0$$

$$\therefore 13t^4 - 10t^2 + 2 = 0$$

$$t = 0 \text{ or } t = \frac{\sqrt{10}}{13}$$

$$d^2 = 13t^4 - 10t^2 + 2$$

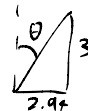
$$d = 0.2774$$

$$\text{(2) } m = 0.12$$

$$a = \begin{pmatrix} 126 \\ 10 \end{pmatrix}$$

$$\underline{s} = \begin{pmatrix} 2t^3 \\ 6t^2+t \end{pmatrix} \quad \underline{v} = \begin{pmatrix} 6t^2 \\ 10t-4 \end{pmatrix}$$

$$\text{When } t = 0.1, \underline{v} = \begin{pmatrix} 2.94 \\ 3 \end{pmatrix}$$



$$\theta = 90 - \arctan\left(\frac{3}{2.94}\right) = 44^\circ$$

$$\text{b) } a = \begin{pmatrix} 6.4 \\ 10 \end{pmatrix}$$

$$F_n = ma = 0.12 \times \begin{pmatrix} 6.4 \\ 10 \end{pmatrix}$$

$$= 1.97$$

$$\begin{aligned} \frac{d}{dt} \sqrt{2t^4 - 6t^3 + 7t^2 + 1} &= \frac{8t^3 - 18t^2 + 14t}{2\sqrt{2t^4 - 6t^3 + 7t^2 + 1}} \\ &= \frac{4t^2 - 9t + 7}{\sqrt{2t^4 - 6t^3 + 7t^2 + 1}} \end{aligned}$$

$$\rightarrow 4t^2 - 9t + 7 = 0$$

$$\text{c) } \frac{10t-4}{6t^2} = 1$$

$$6t^2 - 10t + 4 = 0$$

$$3t^2 - 5t + 2 = 0$$

$$(3t-2)(t-1) = 0$$

$$t = 1 \text{ or } \frac{2}{3}$$